

Indian Real Estate Decision Intelligence: End-to-End Investment & Risk Analytics

1. The Problem Context

The Indian real estate market in 2025 is characterized by high growth but significant **information asymmetry**. Potential homeowners and investors face a "trust gap" due to inconsistent data, registration issue, hidden property ages, and complex pricing structures where "Smart" or "Green" labels are used to justify steep premiums without clear valuation.

2. Problem Statement

This project aims to transform raw, fragmented real-estate data into an Investment Intelligence Engine. Instead of just looking at prices, we analyze risk, space efficiency, and demographic demand to help investors make data-backed decisions.

3. Project Objective

The objective of this project is to develop an **end-to-end data pipeline** (SQL Python Power BI) that transforms raw, messy 2025 property listings into a **Trust & Decision Engine**. This framework will:

- Identify **"High-Risk" vs. "Safe & Verified"** assets using **RERA** and **age** data.
- Find **"Bargains"** where properties are priced 15% below the locality average.
- Analyze **space efficiency** and **parking ratios** to identify high-returns rental assets.
- Map the supply of **"Family-Sized" vs. "Bachelor/Studio"** units to find market gaps.
- Isolate **"Rare Assets" (low-density homes)** in high-density urban markets for long-term land appreciation.

Dataset Summary

(Source: Kaggle datasets; House Property Listings: 2025 Real Estate Data by Pratyush Puri)

- This synthetic dataset comprises residential property listings across multiple Indian metropolitan regions. It includes both training (**train_part1** & **train_part2**) and test (**test_part1** & **test_part2**) dataset.
- Training subset has target prices whereas test subset intended for prediction. So for our analysis **we consider only training subset**.
- We filtered out the columns which are not related to our business problem. We keep following columns for our further analysis:
- Key features:
 - Entity Key: **ListingID** is unique identifier for property listing.
 - Geography: **City** and **Locality** provide hierarchial location; **Latitude** and **Longitude** enable geospatial work.

- Physical Specs: **PropertyType**, **BHK**, **Bathrooms**, **Balconies**, **Furnishing**, **Areas (Super/Built-up/Carpet)**, **Floor/TotalFloor**, **Parking**, **BuildingType**, **YearBuilt** and **AgeYears**, **Facing**.
- Amenities: **AmenitiesCount** aggregates available features.
- Compliance: **IsRERARegistered** and **RERAID** capture regulatory registration metadata.
- Target: **Price_INR** represents listed or transacted value in INR.

Tools Used

- **Data Engineering:** Python (Pandas, NumPy), SQLAlchemy
 - **Database Management:** MySQL (Relational Schema, Window Functions, Joins, Subqueries, CTEs)
 - **Exploratory Data Analysis:** Seaborn, Matplotlib
 - **Business Intelligence:** Power BI (for dashboarding)
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Phase 1: Data Engineering & Feature Engineering

The goal of this phase was to turn raw, messy CSV files into a high-quality "Master Dataset".

1. Data Ingestion

- I developed a Python script using the SQLAlchemy and logging libraries. Instead of a simple copy-paste, I implemented logging to track the success or failure of every row inserted into MySQL. This ensures the pipeline is "Production-Ready."
- To ensure seamless compatibility across the full data stack (**MySQL, Python/Pandas, and Power BI**), I implemented a 'Standardized Naming Convention' using **snake_case**.
- Raw data was first ingested into **SQL** for structured storage and initial cleaning. The processed tables were then exported to **Python (Pandas)** for advanced feature engineering.

2. Data Wrangling

- Instead of treating each dataset separately I combined "train_part1" and "train_part2" table together (using **UNION ALL** Command) so that I get unified master dataset of ~5,000 records. This allowed for deeper exploratory analysis and more robust insights into the Pan-India real estate market.
- To optimize model performance and visual clarity, **non-essential attributes were excluded** during the data transformation phase.

	listingid	city	locality	propertytype	bhk	bathrooms	superbuiltuparea_sqft	carpetarea_sqft	totalfloors	parking
0	HP01139	Bengaluru	None	Villa	3	3.0	1576	1089	1	Covered
1	HP01630	Kolkata	Rajarhat	Apartment	1	2.0	1098	771	6	Basement
2	HP01608	Kolkata	Salt Lake	Apartment	1	2.0	863	611	20	Open
3	HP01979	MMR	Goregaon East	Apartment	2	3.0	1135	817	29	Open
4	HP01399	Hyderabad	Madhapur	Apartment	3	4.0	2662	1778	2	Open
...	...	...	...	...	...	...	...	...	...	...
4723	HP00733	Chennai	Porur	Penthouse	4	5.0	2670	1862	37	Open
4724	HP00905	Hyderabad	Kukatpally	Villa	5	5.0	5077	3527	3	Open
4725	HP00542	Delhi NCR	Noida-Sector 62	Apartment	3	3.0	2092	1417	9	Covered
4726	HP01215	MMR	Vile Parle	Apartment	2	3.0	1083	823	11	Open
4727	HP00443	Chennai	OMR	Apartment	1	2.0	900	598	11	Covered

4728 rows × 19 columns

	buildingtype	yearbuilt	ageyears	amenitiescount	isreraregistered	reraid	latitude	longitude	price_inr
	Bungalow	2018.0	7	3	1	BEN-RERA-9928-63	12.877154	77.656243	17770000
	Mid Rise	2010.0	15	9	1	KOL-RERA-6820-36	22.584035	88.440364	7130000
	High Rise	2006.0	19	6	1	KOL-RERA-5304-84	22.535687	88.416707	5920000
	High Rise	2025.0	0	7	0	None	NaN	72.877790	19140000
	Standalone Building	2010.0	15	6	1	HYD-RERA-7537-95	17.444963	78.431430	20300000
	...	...	...	...	...	...	...	...	...
	High Rise	2008.0	17	9	1	CHE-RERA-1823-72	13.077017	80.344260	26250000
	Standalone Building	2004.0	21	9	1	HYD-RERA-4007-79	17.460544	78.436887	48420000
	Gated Community	1997.0	28	10	0	None	28.776437	77.033101	16880000
	Mid Rise	2010.0	15	7	1	MMR-RERA-7553-86	19.060848	72.800001	17560000
	Gated Community	1999.0	26	9	1	CHE-RERA-2546-26	13.187485	80.307801	7810000

3. Data Cleaning

➤ Duplicates Handling

- When I checked for duplicates in primary key column i.e. **ListingID**, it is found that single **ListingID** is use for two different property i.e having the same ID for a flat in Mumbai and a house in Delhi. **ListingID** should ideally point to one specific home as it is unique identifier.
- The duplicate ListingIDs indicates that the data collection system reused IDs for different properties in different regions. So for finding actual duplicates I used "**Composite Key Duplication Strategy**" i.e. a row is only a duplicate if multiple columns match exactly (e.g., City + Locality + Price_INR + SuperBuiltUpArea_sqft).
- This strategy helps me to conclude that there are no duplicates present in the record even though **ListingID** is same.

➤ **Missing Data Imputation**

some listings have missing details. Specifically, 440 properties lack parking information, and 950 listings do not have a registered RERA ID. Location data (City/Latitude) is missing for approximately 50 properties. So to maintain the integrity of the 4500+ listings, **data imputation** was used instead of removing records. I handled the missing data in following way:

- If the price is missing, we cannot fulfill our goal of "best price" or "affordability." If the city/locality is missing, we can't map it. It is found that 50 cities and 70 localities are missing.
- We cannot impute locality or price value if it is missing but we can **impute the City using the Locality**, provided that the same locality exists elsewhere in the dataset with a city name attached.
- Used cross-reference imputation to recover missing City values based on known Locality mappings, maximizing the dataset's geographic utility.
- For remaining 70 missing Localities, used '**Unknown**' placeholder to preserve price and area data for aggregate city-level analysis.
- A 1-BHK usually has 1 bathroom, while a 4-BHK has 3 or 4. We have imputed the missing values using **median** based on the BHK count.
- We have calculated the **average (mean)** Latitude and Longitude for each city and use those values to fill the gaps. This ensures that even if we don't know the exact street, the property at least appears in the correct city on the map. If we impute it using locality column then it gave us 90% accuracy but since, for remaining 70 missing Localities, used 'Unknown' placeholder so we used "**City-Centroid**" **Imputation Strategy**.
- For filling missing values in 'reraid' column we used two cases:
 - **Case 1:** if a property is not registered, it shouldn't have an ID.
 - **Case 2:** If it is registered but the ID is missing, it should flag it as "ID Not Provided" rather than leaving it as a NaN.
- Some 'year of built' for properties were missing but the age of property was given so we imputed missing values for 'YearBuilt' using logic: "**2025 - age of property**".
- We impute missing values in parking column using "**Mode**" (the most frequent value) of parking for each building type. For example, a "High Rise" is much more likely to have "Basement" or "Covered" parking than a "Standalone Building".

➤ **Data Inconsistencies Audit**

- Converted numerical features (e.g., **YearBuilt** from float to integer) to ensure mathematical accuracy and prevent visual errors in Power BI.
- Transformed **IsRERARegistered** values from binary (0/1) to human-readable format (No/Yes) to improve dashboard accessibility.
- Implemented cross-column consistency checks, ensuring **CarpetArea** does not exceed **SuperBuiltUpArea**, **YearBuilt** cannot be in the future (2026+), and Prices shouldn't be suspiciously less than 1 Lac.

```
df_filtered.info()
✓ 0.0s

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4728 entries, 0 to 4727
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   listingid             4728 non-null   object
1   city                  4728 non-null   object
2   locality              4728 non-null   object
3   propertytype          4728 non-null   object
4   bhk                   4728 non-null   int64
5   bathrooms             4728 non-null   int64
6   superbuiltuparea_sqft 4728 non-null   float64
7   carpetarea_sqft       4728 non-null   float64
8   totalfloors           4728 non-null   int64
9   parking               4728 non-null   object
10  buildingtype          4728 non-null   object
11  yearbuilt             4728 non-null   int64
12  ageyears              4728 non-null   int64
13  amenitiescount        4728 non-null   int64
14  isreraregistered      4728 non-null   object
15  reraid                4728 non-null   object
16  latitude              4728 non-null   float64
17  longitude             4728 non-null   float64
18  price_inr             4728 non-null   float64
dtypes: float64(5), int64(6), object(8)
memory usage: 701.9+ KB
```

```
# Final check for missing values
df_filtered.isnull().sum()
✓ 0.0s

listingid      0
city           0
locality       0
propertytype    0
bhk            0
bathrooms      0
superbuiltuparea_sqft  0
carpetarea_sqft  0
totalfloors    0
parking        0
buildingtype    0
yearbuilt      0
ageyears       0
amenitiescount 0
isreraregistered 0
reraid         0
latitude       0
longitude      0
price_inr      0
dtype: int64
```

4. Feature Engineering

To provide deeper insights, **10+ custom columns (features)** were engineered. *By adding 10+ engineered features, we transformed raw numbers into "Investment Signals" that a buyer can actually understand.*

- **price_per_sqft:** Standardized price unit; allows property price comparison.
- **locality_avg_price_sqft:** The median price of similar properties in the same neighborhood; acts as the "Fair Market Value."
- **price_deviation_pct:** The percentage difference between the listing price and the locality benchmark; identifies bargains or overpriced units.
- **price_category:** Categorizes properties into 4 buckets (Underpriced to Overpriced) based on price quantiles.
- **land_overlap_risk:** Geospatial check that flags multiple listings on the exact same coordinate to prevent fraud.
- **structural_life_stage:** Classifies the building into stages like New, Mid-Life, Aging, and Redevelopment Needed based on age and building type.
- **safety_rating:** A proprietary risk score (Verified, Caution, High Risk) combining RERA status, age, price suspicion, land overlap risk and lifespan of buildings.
- **space_efficiency_grade:** Measures how much of the total area is actually usable (Carpet vs. Super Built-up ratio).
- **family_segment:** Segments properties (Bachelor, Small Family, Large Family) based on BHK and bathroom counts.
- **modernity_index:** Ranks properties by "Smart/Modern" features using amenity counts, parking types, and floor height.
- **utility_balance:** A score determining if the ratio of bathrooms and parking to BHK is sufficient for high rental demand.

- **fsi_category:** Categorized Floor Space Index into three segments (Low, Medium and High FSI) based on the total floors. More number of floors means dense population, because many people sharing same land.
- **low_density_premium:** Flags "Rare Assets"—luxury properties with low building density in high-demand cities.

	price_per_sqft	locality_avg_price_per_sqft	price_deviation_pct	price_category	land_overlap_risk	structural_life_stage	safety_ratings	space_efficiency_grade	family_segment	modernity_index	utility_balance	fsi_category	low_density_premium
0	11275.380711	8724.186239	29.242779	Premium	Inexact Location	New	Verified & Safe	Standard	Large Family Choice	Standard Modern	High-Utility	Low FSI (Spacious/Independent)	Standard
1	6493.624772	7270.668177	-10.687373	Fair Market Value	Low Risk	Mid-Life	Verified & Safe	Standard	Bachelor/Studio	Standard Modern	High-Utility	Medium FSI (Suburban Multi-story)	Standard
2	6859.791425	7488.198983	-8.391972	Fair Market Value	Low Risk	Mid-Life	Verified & Safe	Standard	Bachelor/Studio	Standard Modern	High-Utility	Medium FSI (Suburban Multi-story)	Standard
3	16863.436123	16842.422395	0.124765	Fair Market Value	Low Risk	New	High Risk	Standard	Small Family/Couple	Standard Modern	High-Utility	High FSI (Dense Urban)	Standard
4	7625.845229	7417.763158	2.805186	Fair Market Value	Low Risk	New	Verified & Safe	Standard	Large Family Choice	Basic/Traditional	High-Utility	Low FSI (Spacious/Independent)	Rare Asset
...	...	...	...	...	...	...	...	...	...	...	...	...	...
4723	9831.460674	11191.988276	-12.156264	Fair Market Value	Low Risk	Mid-Life	Verified & Safe	Standard	Large Family Choice	Standard Modern	High-Utility	High FSI (Dense Urban)	Standard
4724	9537.128225	7954.545455	19.895326	Premium	Low Risk	Mid-Life	Verified & Safe	Standard	Large Family Choice	Standard Modern	High-Utility	Low FSI (Spacious/Independent)	Standard
4725	8068.833652	9095.449393	-11.287136	Fair Market Value	Low Risk	Aging	High Risk	Standard	Large Family Choice	Standard Modern	High-Utility	Medium FSI (Suburban Multi-story)	Standard
4726	16214.219760	16086.315013	0.795115	Fair Market Value	Low Risk	Mid-Life	Verified & Safe	Standard	Small Family/Couple	Basic/Traditional	High-Utility	Medium FSI (Suburban Multi-story)	Standard
4727	8677.777778	8634.153513	0.505252	Fair Market Value	Low Risk	Aging	Verified & Safe	Standard	Bachelor/Studio	Standard Modern	High-Utility	Medium FSI (Suburban Multi-story)	Standard

4728 rows × 13 columns

Phase 2: SQL Exploratory Data Analysis (EDA)

I loaded our clean dataset into a MySQL database to solve 10 critical business questions. I used **Window Functions (ROW_NUMBER())** , **Subqueries & CTEs** to extract deep insights.

- **Q1. Find properties that are "Safe" but price is less than the neighborhood average.**

	city	locality	price_inr	price_per_sqft	price_deviation_pct	safety_ratings
▶	Delhi NCR	Gurugram-DLF Phase 3	23690000.00	6077.48	-30.7813	Verified & Safe
	Hyderabad	Jubilee Hills	15790000.00	5704.48	-28.3231	Verified & Safe
	Pune	Kothrud	3840000.00	6541.74	-26.9072	Verified & Safe
	Chennai	Anna Nagar	10010000.00	6818.80	-26.1857	Verified & Safe
	Bengaluru	Hebbal	23950000.00	6701.18	-25.7667	Verified & Safe
	Kolkata	New Town	3160000.00	5693.69	-25.6146	Verified & Safe
	MMR	Bandra West	42840000.00	11903.31	-25.4189	Verified & Safe
	Delhi NCR	Ghaziabad-Indirapuram	3810000.00	6731.45	-25.2061	Verified & Safe
	Delhi NCR	Noida-Sector 62	3940000.00	6758.15	-25.197	Verified & Safe
	MMR	Andheri West	5450000.00	11873.64	-24.9303	Verified & Safe

- **Q2. Find the top 3 properties in every city that offer the best balance of Safety, Modernity, and Price Segment.**

	listingid	city	locality	price_inr	safety_ratings	modernity_index	price_category	total_investment_score	property_rank
▶	HP02155	Ahmedabad	Satellite	5040000.00	Verified & Safe	Ultra-Modern/Smart	Underpriced	30	1
	HP00141	Ahmedabad	Prahlad Nagar	6450000.00	Verified & Safe	Ultra-Modern/Smart	Underpriced	30	2
	HP00235	Ahmedabad	Maninagar	12150000.00	Verified & Safe	Ultra-Modern/Smart	Underpriced	30	3
	HP00068	Bengaluru	Banashankari	3880000.00	Verified & Safe	Ultra-Modern/Smart	Underpriced	30	1
	HP01627	Bengaluru	Unknown	3230000.00	Verified & Safe	Standard Modern	Underpriced	25	2
	HP00998	Bengaluru	Marathahalli	3440000.00	Verified & Safe	Standard Modern	Underpriced	25	3
	HP02177	Chennai	Adyar	5420000.00	Verified & Safe	Ultra-Modern/Smart	Underpriced	30	1
	HP00201	Chennai	Anna Nagar	10010000.00	Verified & Safe	Ultra-Modern/Smart	Underpriced	30	2
	HP01686	Chennai	Porur	12290000.00	Verified & Safe	Ultra-Modern/Smart	Underpriced	30	3
	HP01694	Delhi NCR	Ghaziabad-Indirapuram	16990000.00	Verified & Safe	Ultra-Modern/Smart	Underpriced	30	1
	HP00823	Delhi NCR	Faridabad-Sector 21	3470000.00	Verified & Safe	Ultra-Modern/Smart	Fair Market Va...	25	2
	HP00260	Delhi NCR	Ghaziabad-Indirapuram	3810000.00	Verified & Safe	Standard Modern	Underpriced	25	3
	HP00136	Hyderabad	Gachibowli	16300000.00	Verified & Safe	Ultra-Modern/Smart	Underpriced	30	1
	HP00406	Hyderabad	Madhapur	3480000.00	Verified & Safe	Standard Modern	Underpriced	25	2
	HP00041	Hyderabad	Kukatpally	3810000.00	Verified & Safe	Ultra-Modern/Smart	Fair Market Va...	25	3
	HP01188	Kolkata	Tollygunge	14020000.00	Verified & Safe	Ultra-Modern/Smart	Underpriced	30	1
	HP00156	Kolkata	Salt Lake	2680000.00	Verified & Safe	Standard Modern	Underpriced	25	2
	HP01226	Kolkata	Unknown	3110000.00	Verified & Safe	Standard Modern	Underpriced	25	3
	HP01620	MMR	Thane West	73660000.00	Verified & Safe	Ultra-Modern/Smart	Underpriced	30	1
	HP00233	MMR	Andheri West	5450000.00	Verified & Safe	Standard Modern	Underpriced	25	2
	HP01236	MMR	Vile Parle	7650000.00	Verified & Safe	Standard Modern	Underpriced	25	3
	HP00410	Pune	Baner	3220000.00	Verified & Safe	Standard Modern	Underpriced	25	1
	HP01320	Pune	Aundh	3370000.00	Verified & Safe	Standard Modern	Underpriced	25	2
	HP00682	Pune	Baner	3630000.00	Verified & Safe	Standard Modern	Underpriced	25	3

- Q3. Identify which cities have the highest concentration of "High Risk" properties.

	city	total_listings	risk_percentage
▶	Bengaluru	601	21.96
	Kolkata	555	21.08
	MMR	589	20.88
	Pune	598	20.40
	Delhi NCR	612	19.93
	Chennai	613	19.58
	Hyderabad	572	19.23
	Ahmedabad	588	17.69

- Q4. Calculate the exact "Premium" people pay for modern, high-amenity homes compared to traditional ones in each city.

	city	smart_home_price	traditional_home_price	luxury_premium_pct
▶	MMR	17632.144776	15692.276098	12.36
	Pune	9476.572034	8473.946415	11.83
	Bengaluru	9862.888298	8929.736176	10.45
	Chennai	9450.818923	8778.907407	7.65
	Hyderabad	8388.956250	7834.440532	7.08
	Delhi NCR	9667.446769	9079.733981	6.47
	Kolkata	8076.690417	7641.047333	5.70
	Ahmedabad	5333.444769	5081.547692	4.96

- **Q5. Find properties that have "Standard Efficiency" (moderate space) but are categorized as "Overpriced".**

	city	propertytype	price_category	avg_unit_price
▶	MMR	Villa	Overpriced	22287.675000
	MMR	Apartment	Overpriced	19941.640000
	Bengaluru	Penthouse	Overpriced	13003.930000
	Bengaluru	Villa	Overpriced	12770.770000
	Delhi NCR	Penthouse	Overpriced	12560.420000
	Bengaluru	Row House	Overpriced	12263.590000
	Chennai	Villa	Overpriced	12059.330000
	Pune	Apartment	Overpriced	10975.475000
	Hyderabad	Apartment	Overpriced	10380.730000
	Ahmedabad	Row House	Overpriced	6963.320000
	Ahmedabad	Villa	Overpriced	6846.615000

- **Q6. Does building age significantly impact price in luxury segments vs. budget segments?**

	price_category	structural_life_stage	avg_price_sqft	total_units
▶	Fair Market Value	Aging	9171.46	81
	Fair Market Value	Mid-Life	8842.91	1728
	Fair Market Value	New	9370.13	2237
	Overpriced	Mid-Life	12059.33	1
	Overpriced	New	12697.55	16
	Premium	Mid-Life	11102.97	67
	Premium	New	10785.87	333
	Underpriced	Aging	7104.78	32
	Underpriced	Mid-Life	7662.23	170
	Underpriced	New	7551.32	63

- **Q7. Identify properties with a "High-Utility" signal (good parking and bathroom ratios) that are New build and budget friendly.**

	listingid	city	bhk	structural_life_stage	price_inr
▶	HP02300	MMR	6	New	134200000.00
	HP02101	MMR	5	New	104110000.00
	HP01489	MMR	5	New	103690000.00
	HP02307	MMR	6	New	101120000.00
	HP00929	MMR	5	New	99980000.00
	HP00072	MMR	5	New	96060000.00
	HP01569	MMR	5	New	89950000.00
	HP01182	MMR	5	New	88590000.00
	HP02160	MMR	5	New	83640000.00
	HP00555	MMR	6	New	81380000.00
	HP02354	MMR	5	New	80730000.00
	HP02105	MMR	6	New	79770000.00
	HP01864	MMR	5	New	79710000.00
	HP01306	Chennai	5	New	75960000.00
	HP00789	MMR	4	New	74700000.00
	HP01105	MMR	6	New	74220000.00
	HP01970	Bengaluru	5	New	73290000.00
	HP00833	MMR	5	New	73030000.00
	HP01868	MMR	5	New	72760000.00
	HP01750	MMR	4	New	7000000.00

- **Q8. Which cities are dominated by 'Large Family Choice' segments versus 'Bachelor/Studio'?**

	city	family_segment	property_count
▶	Ahmedabad	Large Family Choice	299
	Ahmedabad	Bachelor/Studio	102
	Bengaluru	Large Family Choice	316
	Bengaluru	Bachelor/Studio	112
	Chennai	Large Family Choice	330
	Chennai	Bachelor/Studio	111
	Delhi NCR	Large Family Choice	330
	Delhi NCR	Bachelor/Studio	98
	Hyderabad	Large Family Choice	306
	Hyderabad	Bachelor/Studio	86
	Kolkata	Large Family Choice	287
	Kolkata	Bachelor/Studio	106
	MMR	Large Family Choice	310
	MMR	Bachelor/Studio	108
	Pune	Large Family Choice	338
	Pune	Bachelor/Studio	103

- **Q9. Find all the 'Rare Assets' (from low_density_premium column) in cities where the majority of listings are 'High FSI'.**

	city	locality	price_inr	fsi_category	low_density_premium
▶	Ahmedabad	Prahlad Nagar	2630000.00	Low FSI (Spacious/Independent)	Rare Asset
	Ahmedabad	Bopal	2650000.00	Low FSI (Spacious/Independent)	Rare Asset
	Ahmedabad	Prahlad Nagar	2740000.00	Low FSI (Spacious/Independent)	Rare Asset
	Ahmedabad	Bopal	2770000.00	Low FSI (Spacious/Independent)	Rare Asset
	Ahmedabad	Bopal	2800000.00	Low FSI (Spacious/Independent)	Rare Asset
	Ahmedabad	Bopal	2910000.00	Low FSI (Spacious/Independent)	Rare Asset
	Kolkata	Tollygunge	2990000.00	Low FSI (Spacious/Independent)	Rare Asset
	Ahmedabad	Vastrapur	3060000.00	Low FSI (Spacious/Independent)	Rare Asset
	Ahmedabad	Bopal	3150000.00	Low FSI (Spacious/Independent)	Rare Asset
	Bengaluru	Whitefield	3170000.00	Low FSI (Spacious/Independent)	Rare Asset
	Ahmedabad	Bodakdev	3200000.00	Low FSI (Spacious/Independent)	Rare Asset
	Kolkata	Jadavpur	3390000.00	Low FSI (Spacious/Independent)	Rare Asset
	Ahmedabad	Bopal	3400000.00	Low FSI (Spacious/Independent)	Rare Asset
	Bengaluru	Whitefield	3540000.00	Low FSI (Spacious/Independent)	Rare Asset
	Ahmedabad	Satellite	3570000.00	Low FSI (Spacious/Independent)	Rare Asset
	Chennai	Adyar	3670000.00	Low FSI (Spacious/Independent)	Rare Asset
	Chennai	OMR	3800000.00	Low FSI (Spacious/Independent)	Rare Asset
	Chennai	Adyar	3830000.00	Low FSI (Spacious/Independent)	Rare Asset
	Delhi NCR	Faridabad-Sa	3850000.00	Low FSI (Spacious/Independent)	Rare Asset

- **Q10. How many "Premium/Luxury" apartments are listed with "None" or "Open" parking?**

	city	luxury_homes_without_secure_parking
▶	Delhi NCR	23
	Kolkata	19
	Hyderabad	18
	MMR	18
	Pune	18
	Ahmedabad	12
	Bengaluru	12
	Chennai	8

The SQL analysis confirmed that markets like **Ahmedabad** offer the best value-for-money, while **Bengaluru** and **Mumbai** carry the highest risk but also the highest potential for rental yield.

Phase 3: Python EDA & Diagnostic Analysis

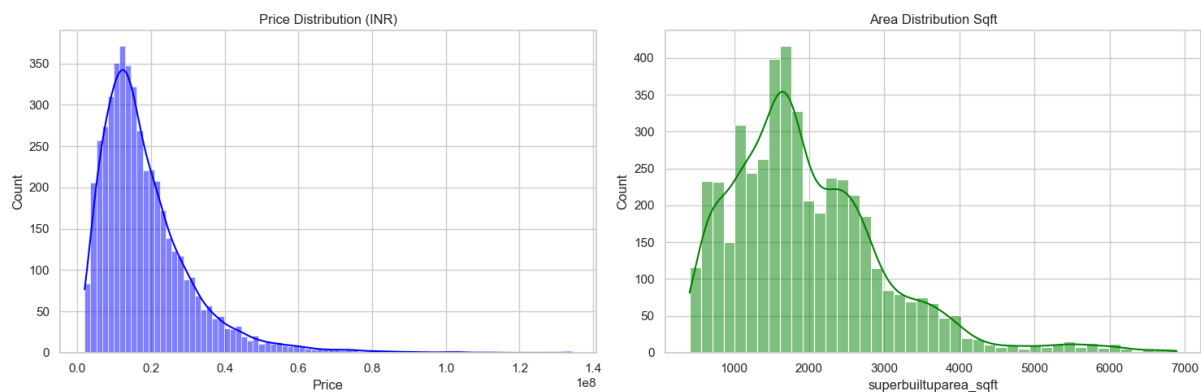
This phase was about "Seeing the Data" and validating our findings. Using Python-based statistical libraries (Seaborn/Matplotlib), I performed an **Exploratory Data Analysis (EDA)** to decode the Indian real estate market.

1. Diagnostic Analysis

Used `df.describe(include='all').T` for summary statistics.

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
listingid	4728	2364	HP01139	2	NaN	NaN	NaN	NaN	NaN	NaN	NaN
city	4728	8	Chennai	613	NaN	NaN	NaN	NaN	NaN	NaN	NaN
locality	4728	60	Noida-Sector 62	104	NaN	NaN	NaN	NaN	NaN	NaN	NaN
propertytype	4728	6	Apartment	2777	NaN	NaN	NaN	NaN	NaN	NaN	NaN
bhk	4728.0	NaN	NaN	NaN	2.551819	1.278891	0.0	2.0	3.0	3.0	6.0
bathrooms	4728.0	NaN	NaN	NaN	3.234137	1.243541	1.0	2.0	3.0	4.0	7.0
superbuiltuparea_sqft	4728.0	NaN	NaN	NaN	1976.857234	1042.352626	406.0	1237.0	1765.0	2512.0	6896.0
carpetarea_sqft	4728.0	NaN	NaN	NaN	1333.160321	673.56898	300.0	854.0	1208.0	1702.0	4185.0
totalfloors	4728.0	NaN	NaN	NaN	12.504653	10.698632	1.0	4.0	9.0	18.0	40.0
parking	4728	4	Covered	2130	NaN	NaN	NaN	NaN	NaN	NaN	NaN
buildingtype	4728	6	High Rise	1547	NaN	NaN	NaN	NaN	NaN	NaN	NaN
yearbuilt	4728.0	NaN	NaN	NaN	2014.143824	7.424854	1985.0	2009.0	2015.0	2020.0	2025.0
ageyears	4728.0	NaN	NaN	NaN	10.856176	7.424854	0.0	5.0	10.0	16.0	40.0
amenitiescount	4728.0	NaN	NaN	NaN	7.022208	1.590023	3.0	6.0	7.0	8.0	12.0
isreraregistered	4728	2	Yes	3778	NaN	NaN	NaN	NaN	NaN	NaN	NaN
reraid	4728	3779	Not Applicable	950	NaN	NaN	NaN	NaN	NaN	NaN	NaN
latitude	4728.0	NaN	NaN	NaN	19.378241	4.972709	12.8006	13.186125	18.64355	22.907	28.7997
longitude	4728.0	NaN	NaN	NaN	77.572772	4.741859	72.4001	73.752575	77.46115	78.59415	88.4998
price_inr	4728.0	NaN	NaN	NaN	18534153.976311	12766028.817688	2000000.0	10070000.0	15310000.0	23330000.0	134200000.0
price_per_sqft	4728.0	NaN	NaN	NaN	9209.157261	3164.994452	3674.54	7487.225	8570.95	9922.5525	24181.95
locality_avg_price_per_sqft	4728.0	NaN	NaN	NaN	9158.608113	3017.239661	4320.99	7687.9	8742.37	9282.05	22426.72
price_deviation_pct	4728.0	NaN	NaN	NaN	0.620882	10.362528	-30.7813	-6.272328	0.0	6.46647	46.8777
price_category	4728	4	Fair Market Value	4046	NaN	NaN	NaN	NaN	NaN	NaN	NaN
land_overlap_risk	4728	2	Low Risk	4658	NaN	NaN	NaN	NaN	NaN	NaN	NaN
structural_life_stage	4728	3	New	2649	NaN	NaN	NaN	NaN	NaN	NaN	NaN
safety_ratings	4728	3	Verified & Safe	3777	NaN	NaN	NaN	NaN	NaN	NaN	NaN
space_efficiency_grade	4728	2	Standard	4579	NaN	NaN	NaN	NaN	NaN	NaN	NaN
family_segment	4728	3	Large Family Choice	2516	NaN	NaN	NaN	NaN	NaN	NaN	NaN
modernity_index	4728	3	Standard Modern	3414	NaN	NaN	NaN	NaN	NaN	NaN	NaN
utility_balance	4728	1	High-Utility	4728	NaN	NaN	NaN	NaN	NaN	NaN	NaN
fsi_category	4728	3	Medium FSI (Suburban Multi-story)	2160	NaN	NaN	NaN	NaN	NaN	NaN	NaN
low_density_premium	4728	2	Standard	3454	NaN	NaN	NaN	NaN	NaN	NaN	NaN

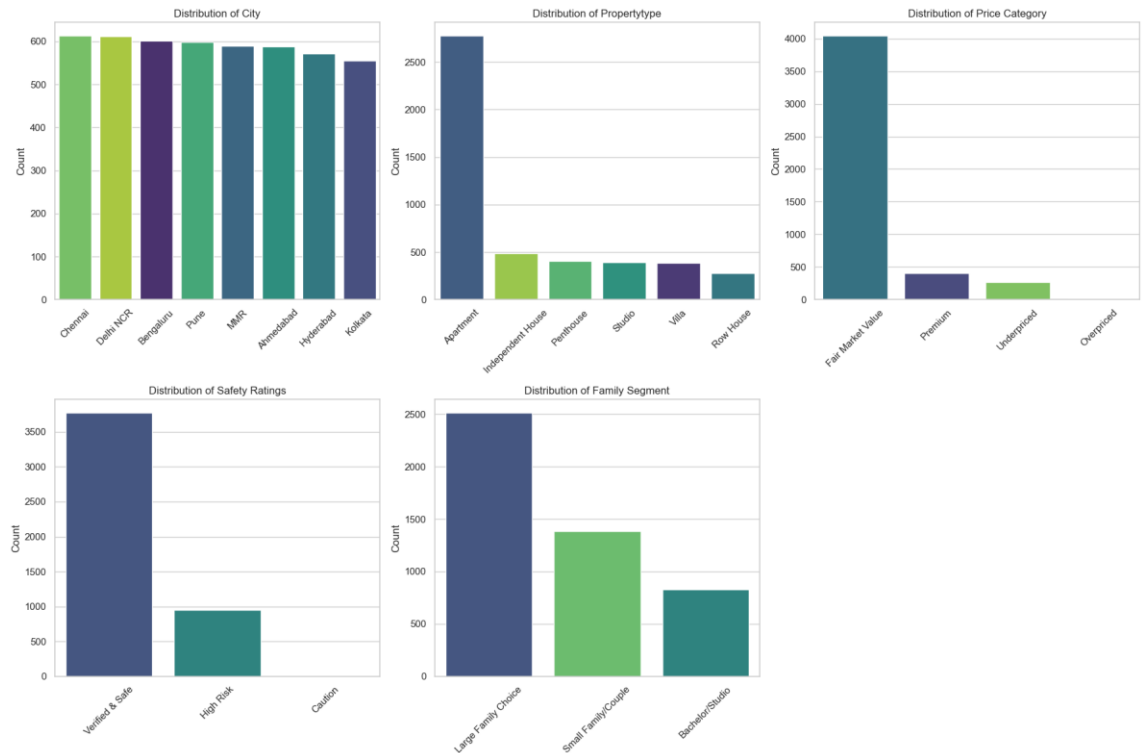
a) **Distribution & Countplots:** I checked the spread of prices and property types.



Key Insights: Market Shape

- **Pricing:** The market is dominated by mid-range housing; high-priced luxury properties represent a small, exclusive niche.

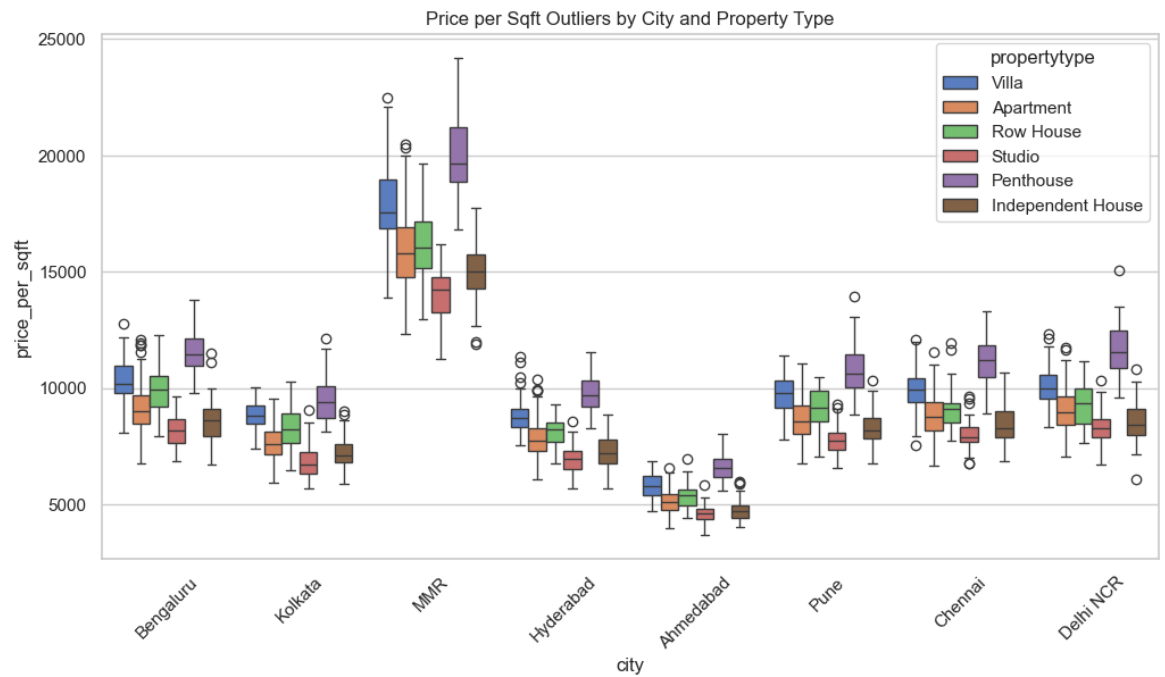
- **Property Size:** Most buyers and developers focus on homes under 4,000 sqft, with very few "mega-sized" properties available.



Key Insights: Top-picks

- **City Presence:** Chennai leads in property availability (more than 600 listings), while Kolkata follows closely behind (~550 listings).
- **Building Trends:** Apartments are the primary choice for urban living (2,500+), vastly outnumbering independent houses (under 500).
- **Market Fairness:** The majority of listings are priced at "Fair Market Value," suggesting a stabilized pricing environment.
- **Risk Levels:** While 75% of properties are "Verified & Safe," there is a significant risk pocket (~900 properties) requiring caution.
- **Target Audience:** The inventory is heavily geared toward large families, with three times more family-sized homes than bachelor studios.

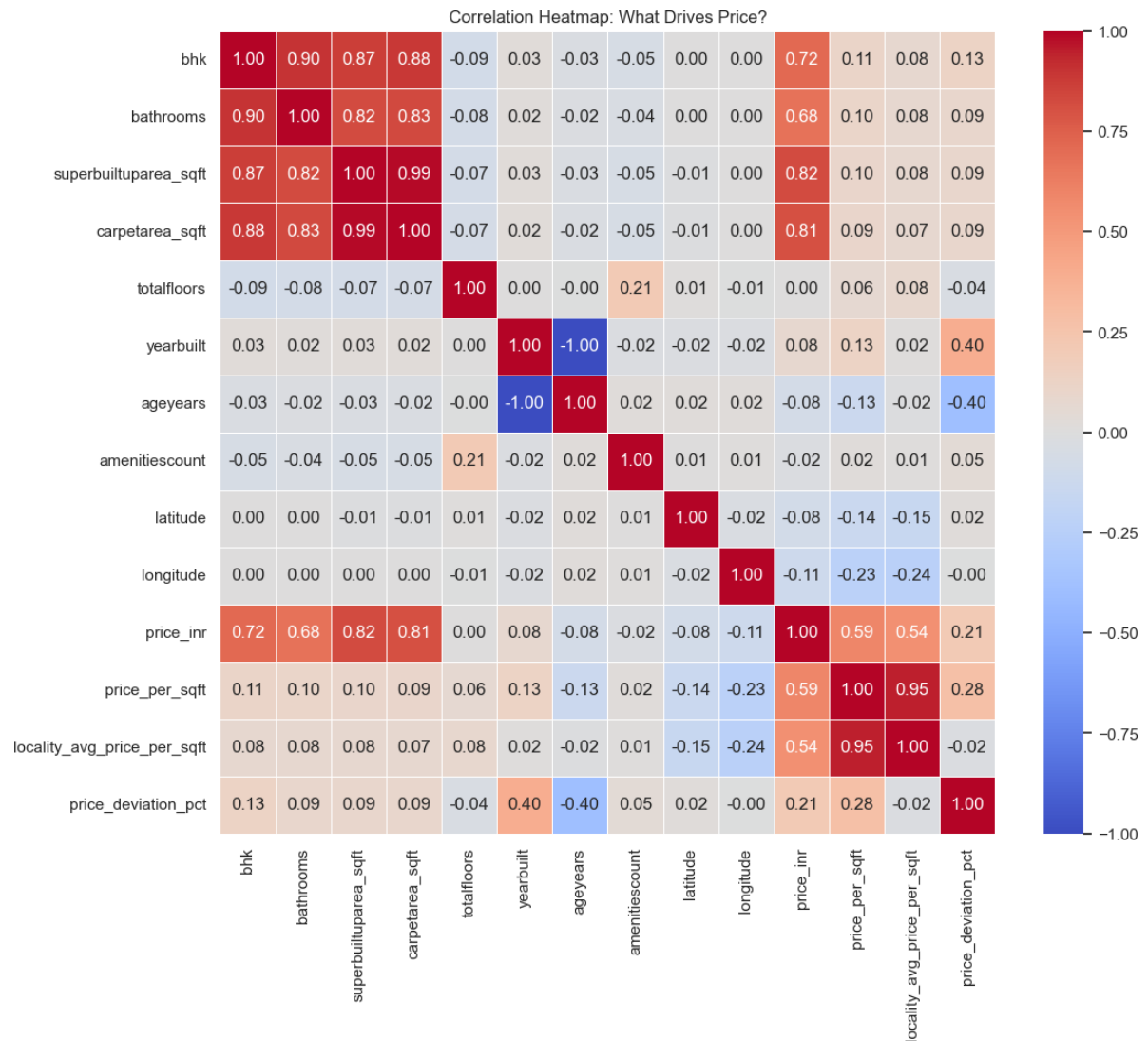
- b) **Outlier Detection:** I identified 230 statistical outliers, representing unique luxury assets that sit outside the typical market range.



Key Insights: Outliers

- **Premium Locations:** Mumbai remains the most expensive market, peaking at ₹22,000 per sqft for luxury penthouses.
- **Budget Entry:** Ahmedabad offers the most affordable entry point for buyers in terms of price per sqft.

c) **Correlation Heatmap:** Interestingly, price and area do not automatically increase with more amenities, suggesting that "location" might be a bigger price driver than "features".



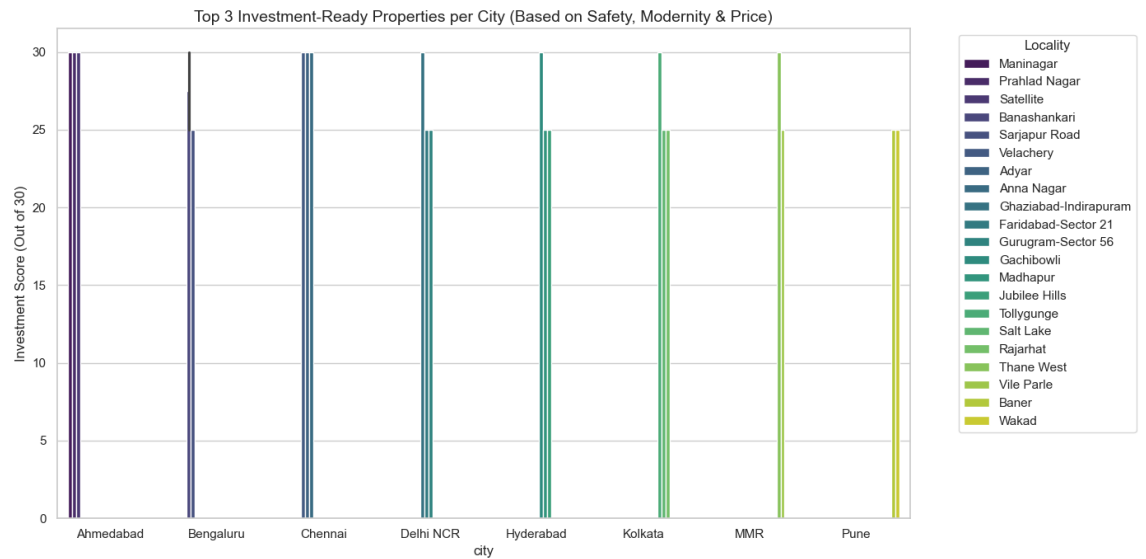
Key Insights: Correlation Heatmap

- **Standard Planning:** There is a nearly perfect match between the number of bedrooms and bathrooms (0.90), showing standardized architectural planning.
- **Age vs. Value:** As building age increases, the price deviation tends to drop (-0.40), confirming that newer buildings command a higher market premium.

2. Visualizing the Questions

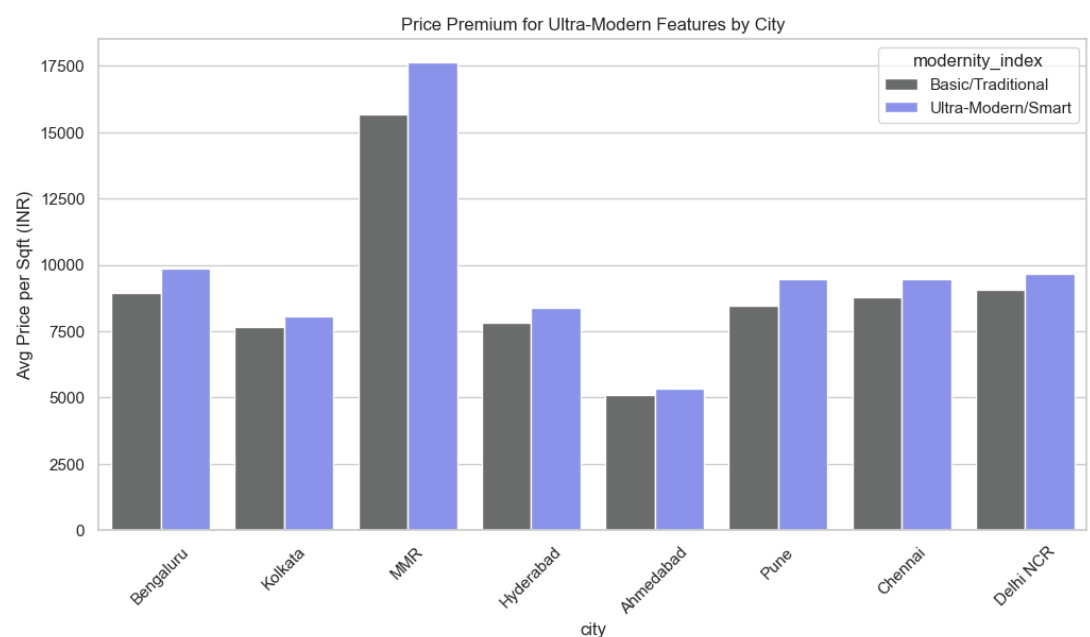
I recreated the SQL results into visual charts (Bar charts, Scatter plots) to provide **Conclusion & Recommendations** for each.

- **Q1. Top 3 properties in every city that offer the best balance of Safety, Modernity, and Price Segment.**



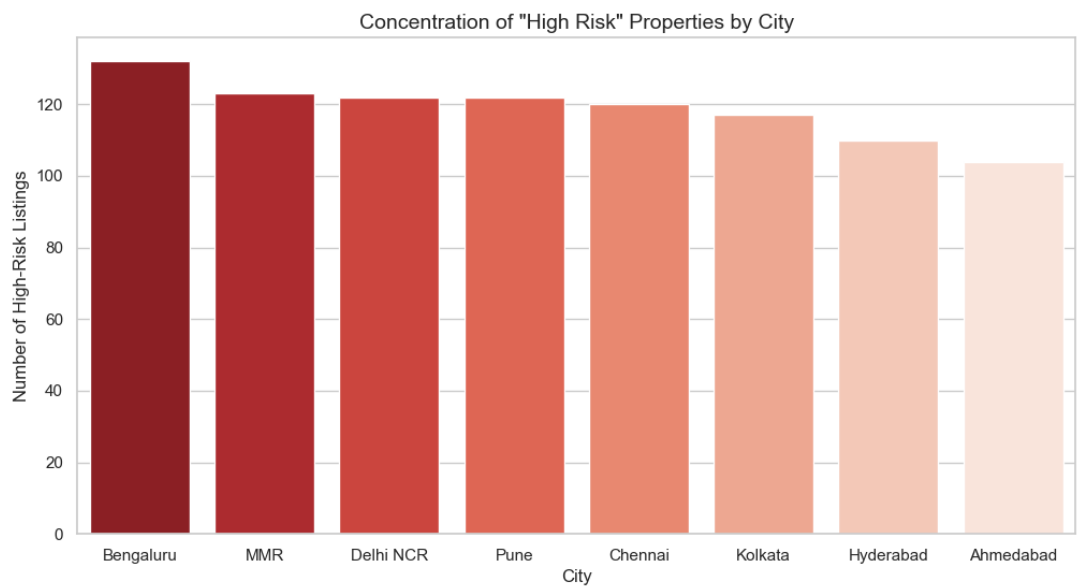
- **Conclusion:** Ahmedabad (Satellite), Bengaluru (Banashankari), Chennai (Adyar), Delhi NCR (Ghaziabad-Indirapuram), Hyderabad (Gachibowli), Kolkata (Tollygunge), MMR (Thane West) and Pune (Baner) offers the highest "Investment Value" due to combination of modern amenities, legal safety, and lower pricing.
- **Recommendation:** Investors looking for low-risk, modern features and budget price segment should prioritize these localities.

▪ **Q2. Find how much extra people pay for "Ultra-Modern" features vs. "Basic" homes across different cities.**



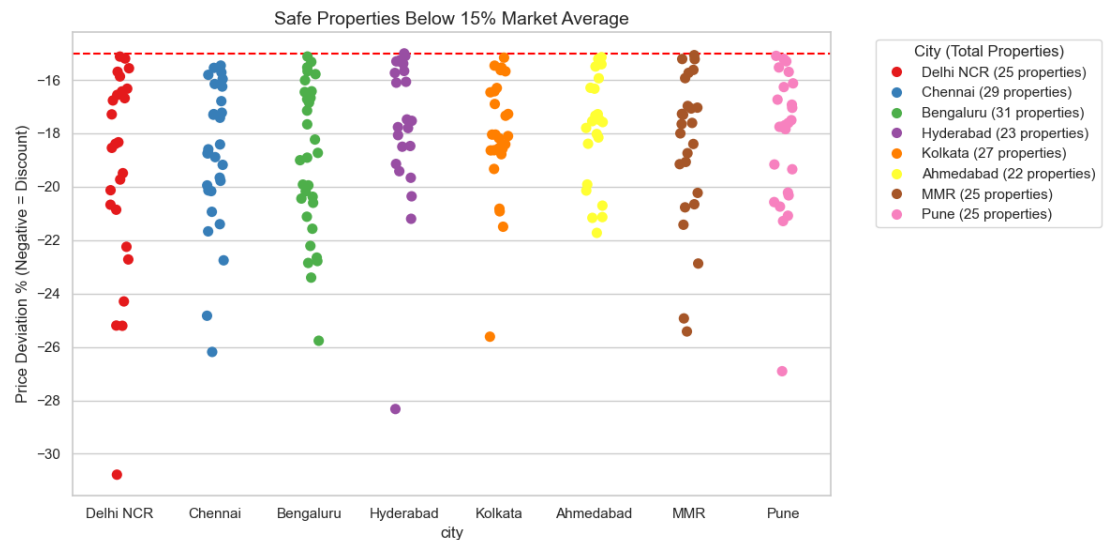
- **Conclusion:** Mumbai's "Basic" homes are priced higher than the "Ultra-Modern" homes of almost every other city. Ahmedabad offers "Smart Luxury" at 1/3rd the cost of Mumbai.
- **Recommendation:** Developers in Ahmedabad can afford more "Smart Home" features to attract luxury buyers, whereas in Mumbai, developers should focus on "Space Optimization" due to already higher prices.

▪ **Q3. Identify which cities have the highest concentration of "High Risk" properties.**



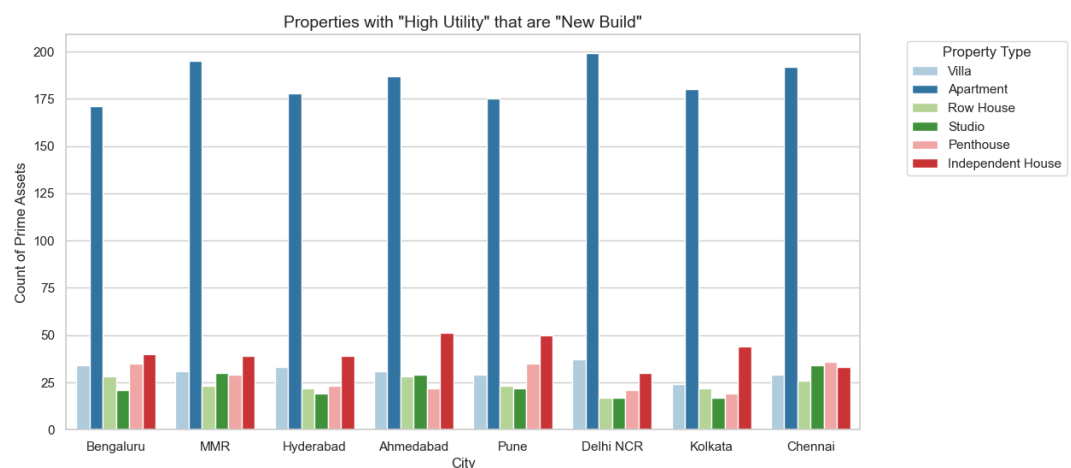
- **Conclusion:** Bengaluru and Mumbai have the highest count of "High-Risk" listings, even though they are tech/finance hubs. The reason for this is probably that they do not have RERA registered or the property is very old.
- **Recommendation:** Buyers looking for a safe home should first check whether the property is RERA registered and the age of the property they are paying for. Developers should prioritize buyer safety first over modern features.

▪ **Q4. Where the safest, most discounted properties are located?**



- **Conclusion:** While Bengaluru has the most bargains, the deepest bargain (-30.78% deviation) is in Gurugram (DLF Phase 3) locality in Delhi NCR.
- **Recommendation:** Real estate buyers should target Gurugram-DLF Phase 3 but double check it before buying property at this price deviation because it may indicate massive market mispricing.

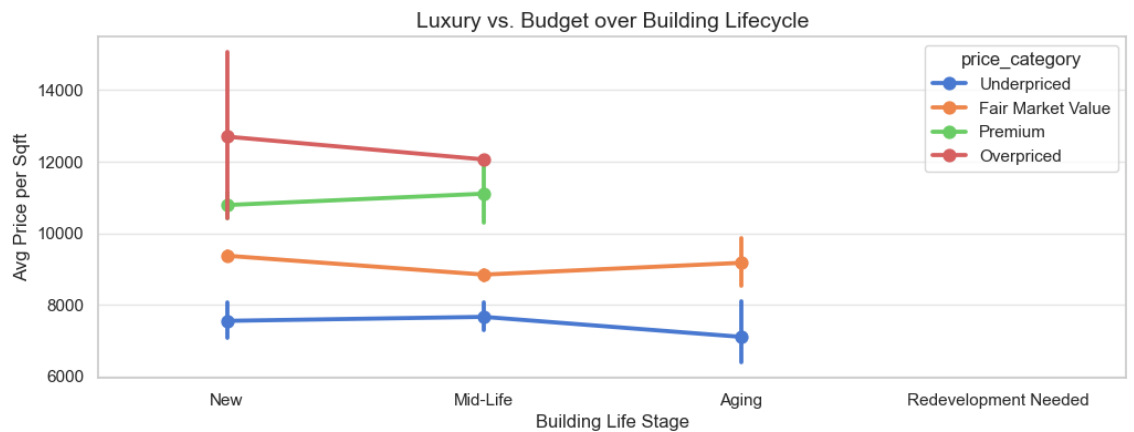
■ **Q5. Identifying properties with a "High-Utility" signal (good parking/bathroom ratios) that are New Builds.**



- **Conclusion:** Mumbai and Delhi lead in "prime assets." The high number of newly built apartments with strong utility ratios indicates they are specifically designed for the rental economy.

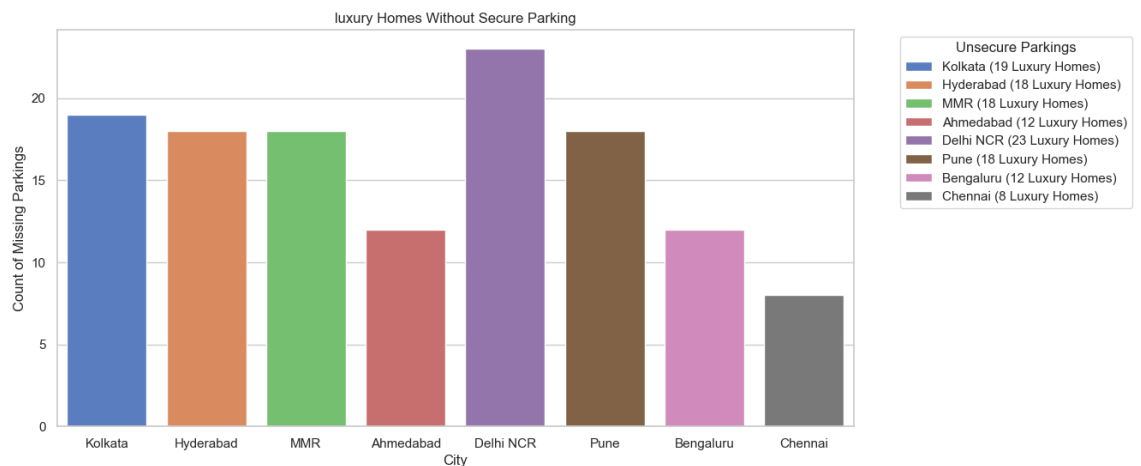
- **Recommendation:** If you want steady money from rent, consider apartments in Delhi and Mumbai. There aren't enough standalone houses in these cities to get profitable business out of them.

▪ **Q6. Does building age significantly impact price in luxury segments vs. budget segments?**



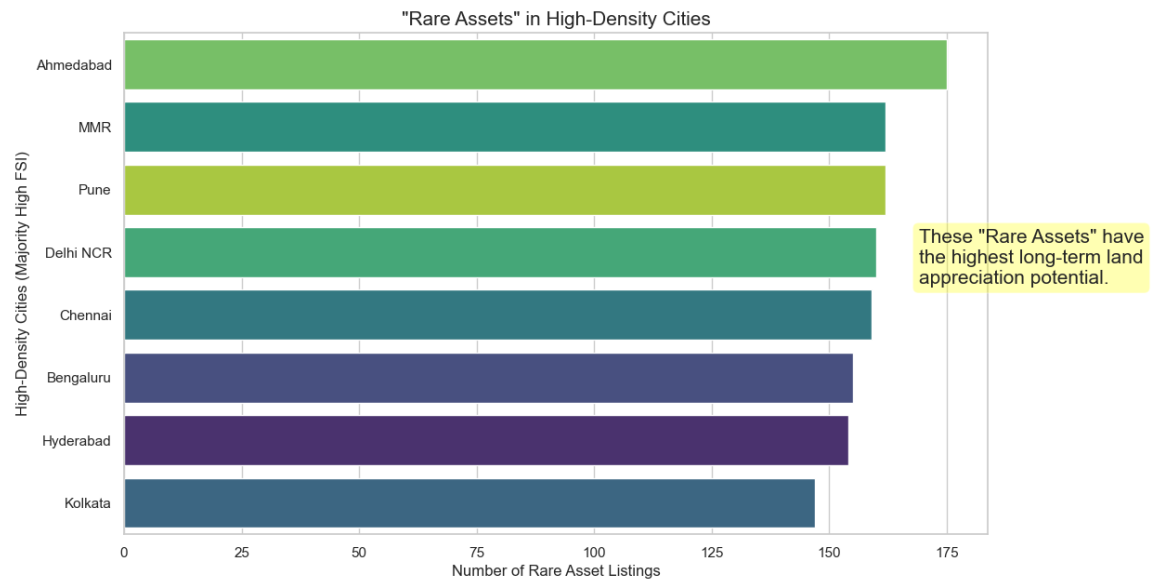
- **Conclusion:** Luxury and Premium segments do not exist in the "Aging" category. Value drops sharply once a building moves past "Mid-Life."
- **Recommendation:** The right time to resale a property is just before a building hits the "Aging" stage. Once a property hits the "Aging" stage, buyers often only care about the land, not the building. They won't pay a premium for the structure because it's outdated.

▪ **Q7. How many "Premium/Luxury" homes are missing secure parking?**



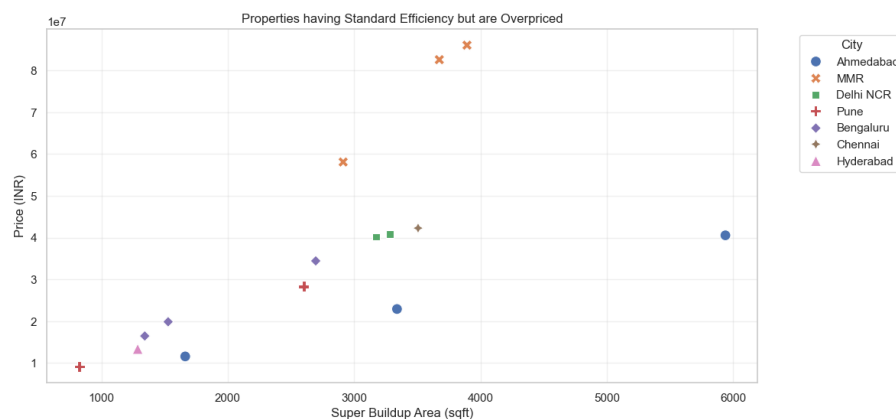
- **Conclusion:** In Delhi NCR 23 premium homes lack secure parking. Chennai has the best alignment of luxury and infrastructure.
- **Recommendation:** Developers should prioritize "Covered Parking" because "Open" or "Unassigned Parking" carries significant disadvantages related to vehicle protection, security, convenience, and legal standing.

- **Q8. Find all the 'Rare Assets' (from low_density_premium column) in cities where the majority of listings are 'High FSI'.**



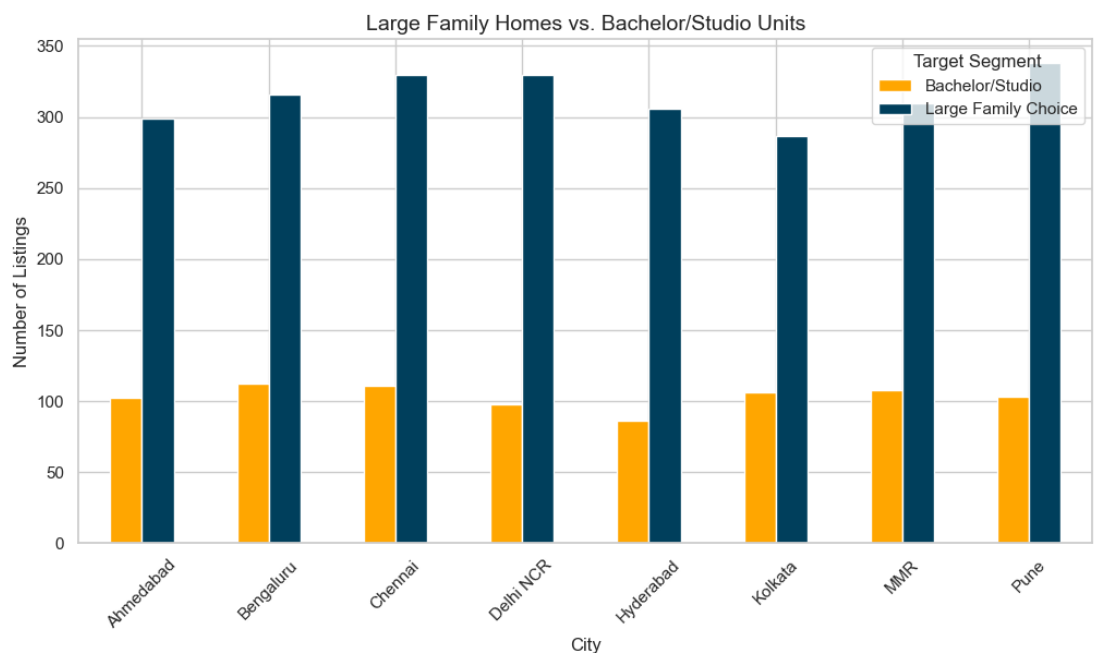
- **Conclusion:** Ahmedabad and MMR offers the most 'Rare Assets' (Low Density) in areas that are otherwise crowded. Because these spacious homes are so hard to find in those cities, they are incredibly valuable.
- **Recommendation:** Buy these properties and keep them in your family for a long time. These will make a lot of money when resold.

- **Q9. Find properties that don't offer "High Efficiency" but are still charging a premium.**



- **Conclusion:** Mumbai has the most "Overpriced & Inefficient" properties. Luxury property in Ahmedabad offer massive 6,000 sqft areas but are still it is inefficient property.
- **Recommendation:** Buyers should use "Space Efficiency" as a negotiation tool. If a 6,000 sqft Ahmedabad villa is "Standard Efficiency," then buyer is paying for dead space (balconies/lobbies) rather than living area.

▪ **Q10. Which cities are dominated by 'Large Family Choice' segments versus 'Bachelor/Studio'?**



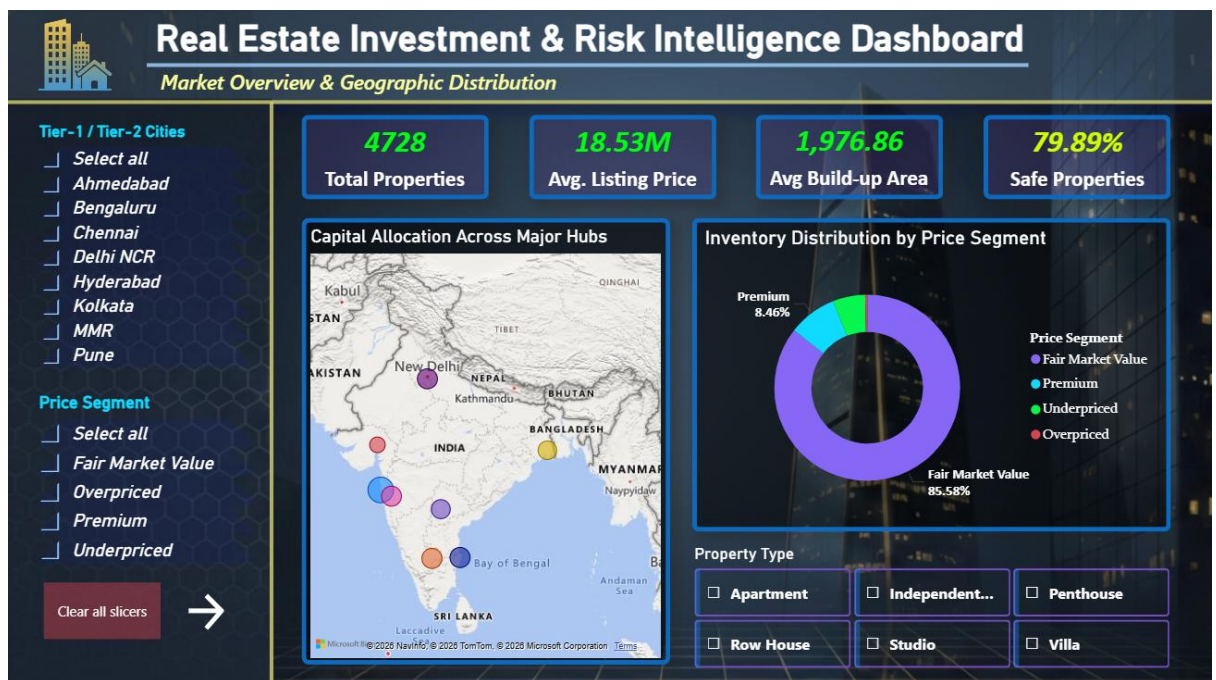
- **Conclusion:** The Indian market is overwhelmingly "Family-First." Bachelor/Studio apartments are a neglected minority.
- **Recommendation:** Cities like Bangalore, Hyderabad, or Pune where thousands of young people move for jobs, there is a huge business opportunity to build small studio apartments and shared living spaces in IT/Tech cities.

Phase 4: Interactive Storytelling (Power BI)

The final phase was building a decision-making tool for non-technical users.

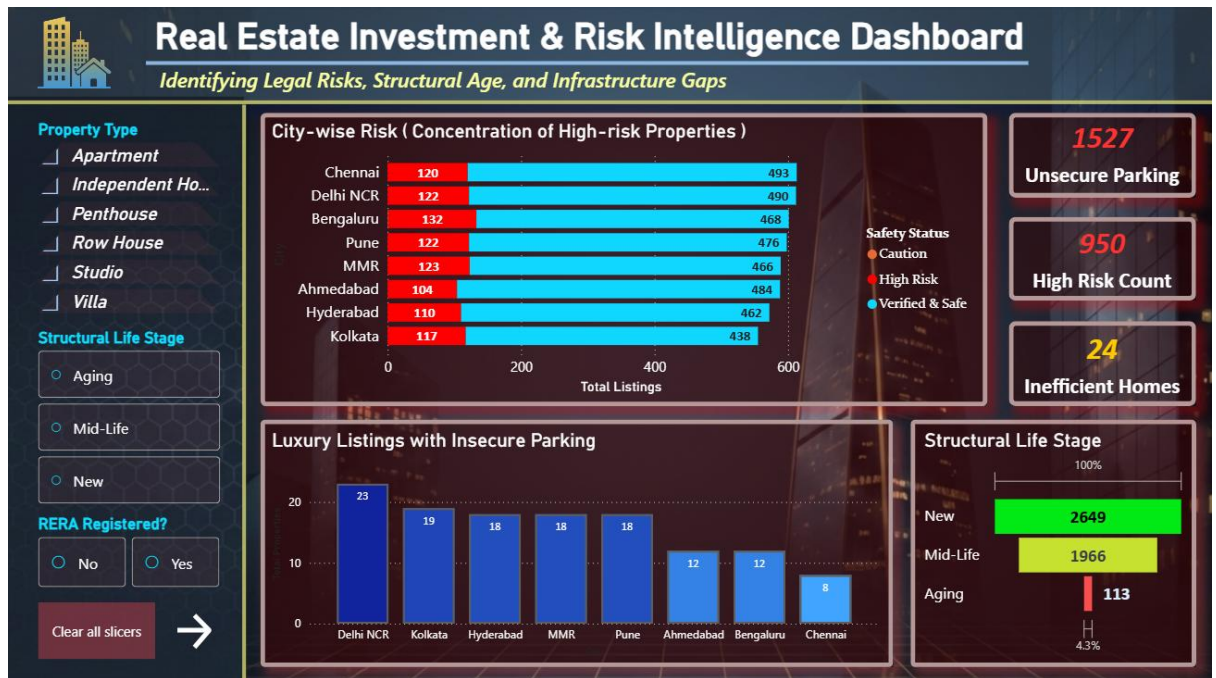
❖ Page 1: Market Overview

- **Map Visual:** Shows capital allocation across major hubs.
- **Donut Chart:** Shows how much of the market is "Budget" vs. "Luxury."
- **KPI Cards:** Instant view of "Total Listings", "Average Prices", "Average Area" & "Total Safe Properties".



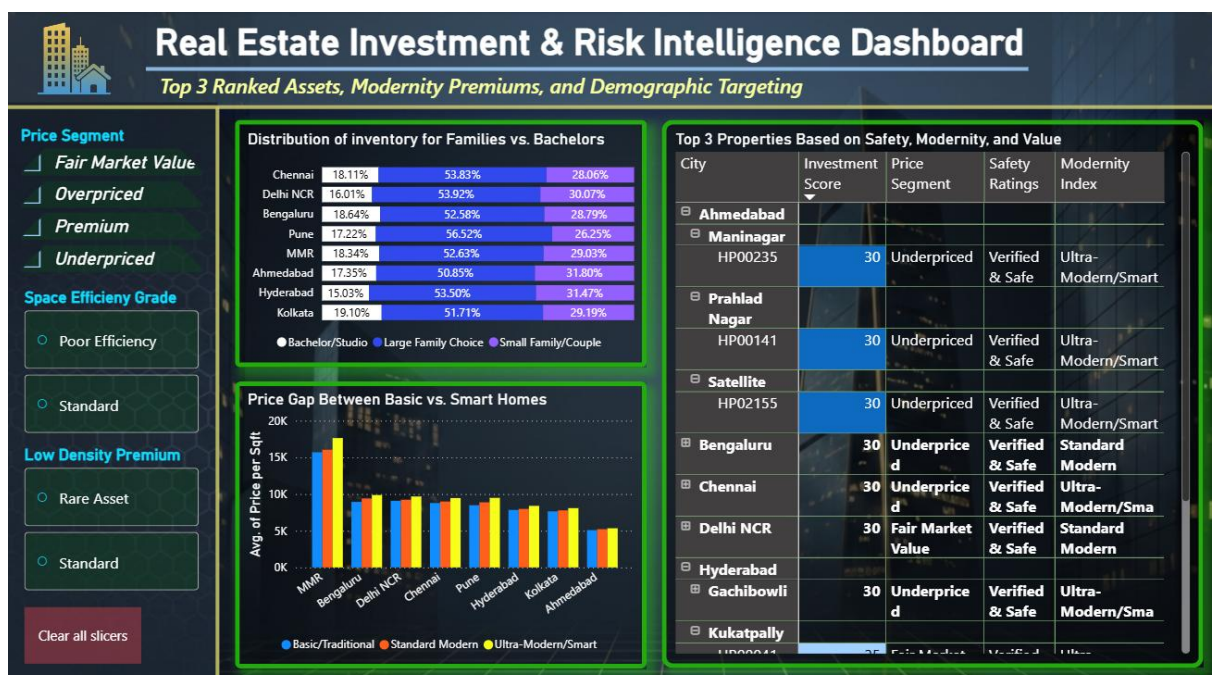
❖ Page 2: Risk & Lifecycle

- **Safety Bar Chart:** Visualizes "High Risk" properties per city.
- **Life Stage Analysis:** Shows how many listings are "New", "Mid-Age" & "Aging".
- **Parking Gap:** Highlights luxury homes that lack secure parking.



❖ Page 3: Investment Strategy

- **The Matrix:** Shows the “Unique Top 3 Properties” for every city.
- **Modernity Premium:** Shows how much extra people pay for "Ultra-Modern" features.
- **Demographic Targeting:** Shows distribution of inventory for “Families” vs. “Bachelors”.



Final Project Conclusion & Final Recommendations

The Indian market is "**Family-Focused**" and "**New-Build**", but riddled with **hidden risks**. There is a massive opportunity in Tier-2 cities like Ahmedabad, Pune because you can get safer, newer, and more modern homes there for much less money than in crowded major hubs like Mumbai or Bangalore.

Final Recommendations:

- **For Investors:** Target "Rare Assets" in high-density cities for long-term land value.
- **For Developers:** Focus on "Bachelor/Studio" units in IT hubs to fill a massive supply gap.
- **For Buyers:** Use the "Safety-First" filter in Bengaluru/Mumbai to avoid the 20% high-risk listings.